

NODE.JS

Digital Notebooks: File System & JSON [cite: 280-282]

Learn to save your code's memory forever! [cite: 283]

Planrs Academy

Curriculum Design Series | Grade 6

The Memory Problem [cite: 286]

Variables are like things we hold in our hands. When we stop working or close our program, they vanish! [cite: 287]

Did you know? [cite: 288]

Your computer's RAM is like a brain that forgets everything when it goes to sleep. To keep our data safe, we need a better way! [cite: 288, 289]

How can we save our notes forever, even after we close our program? [cite: 289, 290]

The Digital Almirah [cite: 291]

The **fs module** (File System) is like a digital cupboard or an Almirah. [cite: 291-293]

Just like your cupboard at home stores clothes safely, the fs module stores your data on the hard drive! [cite: 293]

Why use the Almirah? [cite: 294-297]

- Data Stays Safe [cite: 294]
- Stays saved even when the computer is turned off [cite: 295]
- Permanent Storage that stays forever [cite: 296, 297]

Writing Files with `fs.writeFileSync()` [cite: 302]

We use a special "magic spell" to put data into our digital Almirah. [cite: 302, 304]

```
const fs = require('fs'); [cite: 303]
fs.writeFileSync('notes.txt', 'My first note!'); [cite: 303]
```

This command creates a file called **notes.txt** and saves your text inside it forever! [cite: 303, 304]

Reading Files with `fs.readFileSync()` [cite: 321]

What good is a saved note if we can't read it back? [cite: 322]

Use **`readFileSync`** to retrieve your saved data anytime and display it in your program. [cite: 321, 324]

```
const data = fs.readFileSync('notes.txt', 'utf-8'); [cite: 307]
console.log(data); [cite: 315]
```

Now you can load your saved notes whenever you want! [cite: 322, 323]

Meet JSON: The Recipe Format [cite: 338, 339]

Sometimes, plain text isn't enough. We need structure! [cite: 346]

Think of JSON like a **recipe card** for Maggi or Biryani. [cite: 340]

It has clear labels like "Ingredients" and "Steps" that everyone understands the same way. [cite: 340, 341]

Why JSON? [cite: 342-346]

- Universal format computers everywhere understand [cite: 343]
- Organized data with clear labels [cite: 344]
- Easy to read for both humans and machines [cite: 345]

Packing vs Unpacking [cite: 347, 348]

To store our structured data, we need to pack it up! [cite: 349]

JSON.stringify = Packing a suitcase (Object → String for storage)
[cite: 348, 349]

JSON.parse = Unpacking the suitcase (String → Usable Object)
[cite: 348, 349]

Example: { name: "Asha", grade: 6 } [cite: 349]

Creating a Student Profile [cite: 350, 351]

Let's use our new skills to save a structured student profile! [cite: 351, 352]

```
// 1. Create Object [cite: 354]
const student = { name: "Rahul", rollNo: 15, favSubject: "Math" };
[cite: 355]

// 2. Pack it! [cite: 357]
const jsonStr = JSON.stringify(student); [cite: 358]

// 3. Save to the Almirah! [cite: 359]
fs.writeFileSync('student.json', jsonStr); [cite: 360]
```

Your data is now safely packed and stored! [cite: 358, 360]

Build Your Digital Tiffin Diary! [cite: 361, 362]

Task: Write your lunch menu for the week and save it in **tiffin.json**! [cite: 363]

Sample Menu Data: [cite: 364-369]

- Monday: Idli with Sambar [cite: 365]
- Tuesday: Aloo Paratha [cite: 366]
- Wednesday: Flattened rice dish [cite: 367]
- Thursday: Spicy chickpeas [cite: 368]

Let's save your tasty tiffin notes so you never forget what's for lunch! [cite: 370]

JSON Scavenger Hunt! [cite: 372]

Can you find the mistakes in this broken JSON? [cite: 373, 374]

```
{ name: Rahul, rollNo: 15 favSubject: "Math" } [cite: 374]
```

Hint: [cite: 374]

Look for missing quotes around text and missing commas between items! [cite: 374]

Fix the JSON to make it work properly in your program! [cite: 375]

Recap: .txt vs .json [cite: 376]

.txt Files [cite: 376]

- Simple notes with no structure [cite: 376]
- Just plain text [cite: 376]
- Easy to read but no labels [cite: 376]

.json Files [cite: 376]

- Data with labels and structure [cite: 376]
- Easy for computers to understand [cite: 376]
- Perfect for information with meaning! [cite: 376]

Key Takeaways [cite: 377]

Digital Almirah: [cite: 378] File System lets us save data permanently, keeping notes safe. [cite: 378]

Recipe Format: [cite: 379] JSON is a universal format to store structured data so everyone reads it the same way. [cite: 379]

Pack & Unpack: [cite: 380] We can stringify and parse data easily to keep our notebook safe and ready. [cite: 380]

Your Code's New Superpower! [cite: 381]

"We are giving our code a long-term memory!" [cite: 381]

What will you create next with your Digital Notebook? [cite: 382]

Your programs can now remember everything—notes, recipes, student profiles, and more! [cite: 382]

[Image of a student building a creative digital project]

THANK YOU! [cite: 383]

We hope you enjoyed learning about File System & JSON APIs.
[cite: 383]

Next Session: Coming Soon! [cite: 384]

Keep coding and building your Digital Notebooks! [cite: 384]